Zach Sahlin

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WORK EXPERIENCE

Software Engineer | MAQ Software *June 2024 – Present*

- Developed data pipelines using Databricks, Fabric, and Azure Data Factory, transforming user telemetry data of AI tools into curated datasets enabling data science and reporting for ROI measurements
- Refactored and optimized Spark ML pipeline in Azure Databricks to cut runtime by 95% and reduce monthly refresh delay from 3 days to overnight
- Led a team of 5 engineers to deliver stakeholder defined requirements for SMB AI initiatives
- Proposed and implemented robust outlier filtering logic to resolve the unexpected variability of ROI calculations
- Spearheaded requirement gathering by collaborating with stakeholders and proactively raising issues and concerns in order to provide clear technical specs streamlining development
- Managed multiple projects creating Power BI dashboards with 5,000+ monthly users for analyzing sales data for an enterprise cloud provider
- Leveraged AI tools including Databricks Assistant, GitHub Copilot, and M365 Copilot daily to accelerate integration with a legacy database

Teaching Assistant | Gonzaga University *August 2022 – May 2023*

- Assisted Dr. Yanping Zhang in Operating Systems, Computer Security, and Internet of Things courses
- Critically analyzed, tested, and graded programming and written assignments

EDUCATION

Gonzaga University | Spokane, WA August 2019 – May 2023

BS in Computer Science, Minor in Applied Mathematics

- GPA: 3.7, Magna Cum Laude

PROJECTS

Stock Predictor | Python, TensorFlow, AWS Lambda, AWS EventBridge, Selenium *Spring 2023*

- Designed a model to predict short term prices of stocks
- Created a data pipeline with web scrapers and APIs to collect historical and current data from various sources
- Leveraged AWS S3 to store and retrieve data
- Implemented a CI/CD pipeline with GitHub Actions to test and publish to AWS Lambda
- Developed a web app to display predictions updated daily

Machine Learning for Real Estate | Python, SQL, PyTorch, JS, FastAPI *Fall 2022 - Spring 2023*

- Collaborated with my Senior Design group to apply machine learning to real estate
- Collected, cleaned, and managed data on historical rental prices, crime data, and school scores
- Implemented and trained an LSTM to predict future rent prices
- Improved the REST API to decrease front end load time from 5 minutes to about 5 seconds
- Developed a web app to display historical prices and predictions in a graphical interface

Resume Builder | Python, JavaScript, TypeScript, React, Next.js, MongoDB, LaTeX, AWS Lambda *Fall 2023*

- Created a web app with React and Next.js to allow users to input resume content and manage multiple resumes
- Developed an AWS Lambda function to format a resume in LaTeX and build the resume as a PDF
- Solved Lambda dependency and image size constraints by leveraging AWS ECR

RELATED COURSEWORK

- Algorithms & Abstract Data Structures
- Machine Learning & Intelligent Systems
- Data Science Algorithms
- Database Management Systems
- Linux & DevOps
- Natural Language Processing

TECHNICAL SKILLS

- Languages: Python, SQL, Java, C++
- Tools: PySpark, Databricks, Fabric, DuckDB, Git, Linux, Docker, Jupyter, GitHub Actions, Postgres, AWS Lambda, AWS S3, AWS ECR
- Libraries: TensorFlow, PyTorch, pandas, NumPy, scikit-learn, Matplotlib